

---

**Input:** Specification set  $\mathcal{D}$ ; reference policy  $\pi_{\text{ref}}$ ; rubric weights  $w$ ; group size  $G$ ; temperature  $\tau$ ; nucleus  $p$ ; clip  $\varepsilon_c$ ; KL weight  $\beta$ ; number of iterations  $K$

**Output:** Fine-tuned adapter parameters  $\theta$

```
1  $\theta \leftarrow$  adapter initialization;  $\theta_{\text{old}} \leftarrow \theta$ ;
2 for  $k \leftarrow 1$  to  $K$  do
3   sample a batch  $B \subset \mathcal{D}$  of specifications;
4   foreach  $s \in B$  do
5      $q \leftarrow$  prompt( $s$ ); sample  $o_1, \dots, o_G \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$  with  $(\tau, p)$ ;
6     for  $i \leftarrow 1$  to  $G$  do
7        $c_i \leftarrow \text{PARSE}(o_i)$ ; // cascade of parsers, Section 6.3.5
8       if  $c_i = \perp$  then  $r_i \leftarrow 0$ ; continue;
9        $|\psi_{c_i}\rangle \leftarrow \text{CUDAQSIMULATE}(c_i)$ ; // GPU state vector
10       $r_i \leftarrow \sum_k w_k r_k(c_i)$ ; // Equation (6.2): fidelity, Clifford, topology, T,
        depth
11     $\hat{A}_i \leftarrow (r_i - \text{mean } r) / (\text{std } r + \delta)$  for all  $i$ ; // Equation (6.4)
12     $\theta \leftarrow \theta + \eta \nabla_{\theta} \mathcal{J}_{\text{GRPO}}(\theta)$  over the batch; // Equation (6.5), DeepSpeed ZeRO-2
13     $\theta_{\text{old}} \leftarrow \theta$ ;
14 return  $\theta$ ;
```

---